



Trust Vote: Online Voting using Blockchain

Aim of the Project

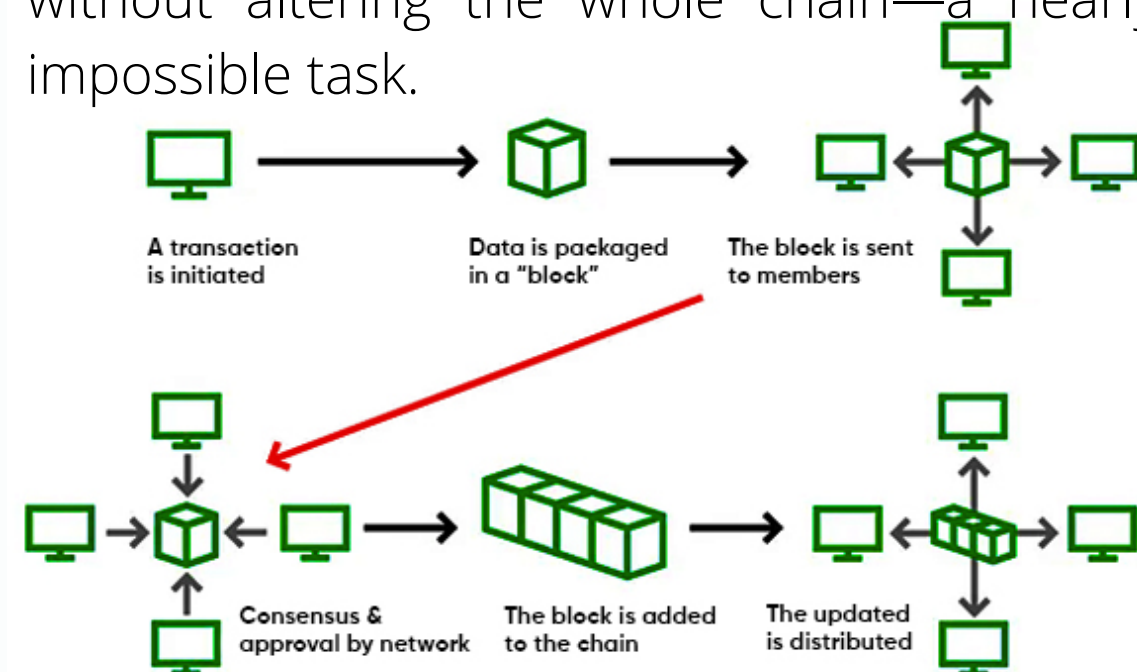
- A blockchain-based e-voting system offers a solution by providing a secure, transparent, and tamper-proof platform for elections.

What is Blockchain Technology ?

Blockchain is a digital ledger that records transactions across many computers in a secure and transparent way. Transactions are grouped into "blocks" and linked together, making them tamper-proof. It's decentralized, meaning no single authority controls it, ensuring security and trust. This technology is used in cryptocurrencies and other fields like voting and supply chains.

How does Blockchain Work ?

- Transactions are grouped into "blocks" with each block linked to the one before it, forming a secure chain.
- Instead of one central place, blockchain is spread across many computers (nodes). Everyone has a copy, so no one can cheat.
- Transactions are checked by the network (using math puzzles or voting) to ensure they're real before being added to the chain.
- Once added, data in a block can't be changed without altering the whole chain—a nearly impossible task.



Proposed System

- **Registration:** Voters register through a form on the web application, and their details are recorded on the blockchain.
- **Login:** Users log in using their Aadhar number, Email and password to access their account.
- **Dashboard:** After logging in, the dashboard displays voting information and options.
- **Voting:** Users select a party from the list of candidates and cast their vote. Only one vote is allowed per account.
- **Email Verification:** After voting, an email verification is sent, and the vote is securely recorded on the blockchain.
- **Secure Login:** The system ensures secure login, preventing unauthorized access and vote tampering.
- **Admin Panel:** Admins can add new candidates and view registered users' details in the admin panel.

SIGN-IN

VOTE PAGE

#	Name	Party Name	Votes
1	Narendra Modi	Bharatiya Janata Party	1
2	Rahul Gandhi	Indian National Congress	1
3	Arvind Kejriwal	Aam Aadmi party	1
4	Mamata Banerjee	All India Trinamool Congress	0

ADMIN PANEL

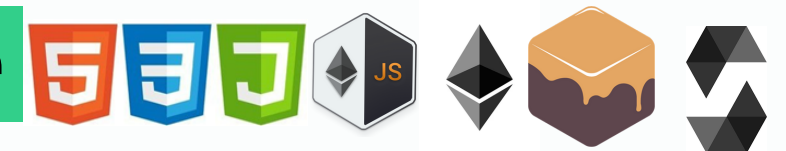
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RESULT PAGE

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Architecture



- Frontend: The user interface is built using **HTML, CSS, and JavaScript**, allowing users to interact with the application, register, log in, and vote.
- Backend: The server-side logic is handled by **Node.js** and **Express.js**, managing authentication, user data, and communication with the blockchain.
- Database: **MongoDB** stores non-sensitive user information like login credentials, while sensitive data (votes) is stored securely on the blockchain.
- Solidity & Smart Contracts: **Solidity** is used to write **smart contracts**, which are self-executing programs that handle tasks like voting, vote recording, and result tallying on the Ethereum blockchain.
- **Web3.js, Truffle, Ganache & MetaMask:** Web3.js connects the frontend to the Ethereum blockchain, Truffle is a development framework for smart contracts, Ganache provides a local blockchain for testing, and MetaMask allows users to interact with the blockchain through their browser.

Result

This project creates a secure online voting system using blockchain, ensuring transparent and tamper-proof voting. Users can register, log in, vote, and receive email verification. Results are displayed on the dashboard, and an admin panel manages candidates and users. The system ensures trust and security in elections.

Industry Mentor

ASHA KADAM